BAOFENG (JUSTIN) JIA

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EDUCATION

Doctor of Philosophy – Molecular Biology and Biochemistry and SFU-UBC NSERC-CREATE Bioinformatics Graduate Program Sept. 2017 – Feb. 2023
Simon Fraser University, Burnaby, BC
- Dissertation title: Advancing Antimicrobial Resistance (AMR) Surveillance: Characterizing AMR Mobility and Improving AMR Prediction from Metagenomics Data

Honours Bachelor of Science – Biochemistry Coop Sept. 2012 – Apr. 2017
McMaster University, Hamilton, ON

SKILLS

*omics data analysis (genomics, transcriptomics, proteomics, and metabolomics). Next generation (Illumina) and third generation (ONT Nanopore) sequencing. Data Science. Correlation and machine learning based biomarker identification. Machine learning and artificial intelligence-based modelling and prediction. Omics tool development. Analysis workflows integration. Mathematical and statical modelling. DNA variant identification. Metagenomics, including amplicon and whole genome shotgun sequencing. Programming languages (Python, R, C#). Version control (Git, GitHub). Cloud computing (UNIX, AWS, docker). Workflow management tools (Galaxy, Nextflow). Database (MySQL).

EXPERIENCES

Project Manager – iMicroSeq and ChuNet Initiatives Apr. 2025 – Present
Simon Fraser University, Burnaby, BC
- Coordinate a national network of over 80 researchers, experts, and knowledge users across academia, government, industry, and Indigenous communities to bridge silos in Canadian wastewater and water-based genomic monitoring.
- Contribute to the creation of an environmental sequence *data specifications* and the *iMicroSeq database* while leading the data curation, and integration efforts to enable inclusive integration of environmental sequence data across Canada.
- Oversee development of bioinformatics tools to enhance wastewater and water-based public health monitoring and environmental genomic data interoperability.
- Engage Northern, Rural, Remote, and Isolated (NRRI) communities to expand eDNA monitoring capacity, supporting community-led decision making grounded in both Indigenous and western knowledge systems while upholding data sovereignty.
- Manage project timelines, scientific reporting, and risk mitigation for multiple initiatives, ensuring alignment with network objectives, partner expectations, and funding agency requirements.

Postdoc, Brinkman Lab – Bioinformatics and Microbiome Analysis Feb. 2023 – Apr. 2025
Simon Fraser University, Burnaby, BC
- Bioinformatics lead and co-coordinator of the Canadian Coronavirus Variants Rapid Response Network (CoVaRR-NET), Computational Analysis, Modelling and Evolutionary Outcomes (CAMEO) initiative.
  - Performed data analysis, coding, visualization, and maintenance of the CoVaRR-Net Duotang notebook for COVID19 genomic epidemiology and mathematical modelling to allow prediction of variants' rate of spread and their potential impact on health care systems.
  - Investigation of emerging bioinformatics and computational tool needs, and subsequent workflow construction for detection of SARS-CoV-2 in wastewater metagenomic data.
  - Communication of key findings from SARS-CoV-2 genomic analysis across CoVaRR-Net pillars and to provincial health to assist in public health decision making.

- Project Analytics for CHILD Cohort study, Brinkman Lab
    - Identification of bacterial biomarkers for medication use on the gut microbiota and modelling the impacts of medication practices on infant microbial dysbiosis.
    - Improved the curation guidelines for the ontological mapping of plain-text reason for medication use to ontology databases, which allowed expanded queries and potential machine learning based applications.

PhD Thesis, Brinkman Lab Sept. 2017 – Feb. 2023
Simon Fraser University, Burnaby, BC

- Collaborated and performed multi-omic data analysis, including transcriptomics, proteomics, and metabolomics. This led to the identification and subsequent *in vitro* investigation of the clinical and functional significance of the interaction between two key metabolism genes in breast cancer.
- Applied natural language processing-based machine learning/artificial intelligence methodologies to predict AMR genes and Genomic Islands from genomics data. This resulted in a preliminary tool that showed improved accuracies compared to existing tools.
- Generated metagenomic microbiome data using Illumina and Oxford Nanopore technologies and applied correlative and machine learning based methods to identify potential biomarkers of contaminant exposure in the St. Lawrence Estuary belugas that can be validated *in vitro*.
- Performed large scale analysis of existing publicly available genomics data to gain insight into mobility trends for antimicrobial resistance (AMR) gene. This allowed the identification of genes with high risk for environmental transmission that may be used for public health risk assessment.
- Evaluated and highlighted the pitfalls of current genomic methods for identification of AMR genes in environmental metagenomic datasets, and then developed a new read-based AMR prediction workflow that can be used to provide a more comprehensive prediction of environmental resistome.

Bioinformatics Consultant July 2021 – Feb 2022
the(sugar)science, thesugarscience.org

- Led the genomics-driven data analyses for the identification of early biomarkers of diabetes.
- Utilized *in silico* genomics analysis to provide evidence that human beta cell GAD65 gene and certain gut bacterial GAD gene showed sequence and motif similarities, and overlap with human T-cell receptor epitopes, which may lead to the immune destruction of beta cells in Type 1 Diabetes.

Program Advisory Committee Member May 2020 – Oct. 2022
Langara College, Vancouver, Canada

- Member of the undergraduate Bioinformatics program advisory committee (PAC) at Langara College.
- Provided feedback and guidance relating to the initial and ongoing development of the bioinformatics program at Langara College that led to an 4x increase in student registration year-over-year.

Website Administrator
MicrobiomeCanada.ca Sept. 2017 – Present

- Led the design, generation and revision of webpages and posts that resulted in a large, and public, repository of microbiome researchers to facilitate cross-Canada discussions.
- Managed the microbiomecanada.ca webservers and maintenance of software operations.
- Performed routine web security audits (access rights, web traffic analysis, etc.)

bcbioinformaticsgrad.ca Mar. 2022 – June 2022

- Led the update and redesign of webpages and posts which resulted in an improved website that provided up to date information to prospective students who want to enter the bioinformatics program.

Bioinformatics Internship
BC Center for Disease Control, Vancouver, BC, May 2018 – Aug. 2018

- Produced an easy to use, user-friendly workflow with a web interface using galaxy to construct phylogenetic trees with epidemiologically useful information from raw sequencing reads. This includes identifying genomics information such as sequence typing, AMR profile prediction and plasmid transmission information.

Research Assistant
Austin Lab, McMaster University, Hamilton, Ontario May 2016 – December 2016

- Studied the effect of TDAG51's Poly-PQ domain on apoptosis using plasmid transfection and quantification techniques like LDH release assay, TUNEL assays and Annexin V staining in order to understand TDAG51's prominent role in diabetes and atherosclerosis.

Senior Thesis, Larche Lab, McMaster University September 2015 – April 2016

- Developed a bioinformatics tool to predict >800 T cell epitopes of 28 cockroach allergens to be potentially formulated into an immunotherapy product for the treatment of cockroach allergy.
- Developed an *in vitro* protocol to determining the accuracy of the *in silico* predicted T cell epitopes using a ligand (epitope)-receptor (MHC Class II) competition binding assays and ELISA.

Comprehensive Antibiotic Resistance Database (CARD) BioCurator
McArthur Lab, McMaster University, Hamilton, Ontario January 2015 - September 2015

- Maintained and Updated the CARD (card.mcmaster.ca), resulting in one of the most robustly curated databases with over 3,000 known resistance genes, gene targets and antibiotics.
- Designed and created the new CARD database schema using PostgreSQL, termed 'Broad Street', resulting in a database 1/5th of the original size and allowing the addition of new prediction models (e.g. indel mutations) for curated data, making it future-proof.
- Collaborated with a team of two software developers to create the Resistance Gene Identifier Ver. 3, an upgrade of the bioinformatics tool that uses CARD's data to predict resistance profiles of any bacteria genome. The upgrade allowed for separation of software from the database allowing for independent software development and resistance data curation.

SELECTED PUBLICATIONS

1. **Jia B**, Alcock BP, Raphenya AR, Spence J, Maguire F, Beiko RG, McArthur AG, Bertelli C, Brinkman FSL. (2026). Analysing antimicrobial resistance mobility patterns using a diverse dataset of over 8000 bacterial species. BioRxiv. DOI: 10.64898/2026.03.03.709307
2. Gill EE, **Jia B,** Murall CL, Poujol R, Anwar MZ, John NS, et al (365 authors total, including 2 consortiums). (2024). The Canadian VirusSeq Data Portal and Duotang: open resources for SARS-CoV-2 viral sequences and genomic epidemiology. Microbial Genomics. 10(10) DOI:10.1099/mgen.0.001293
3. **Jia B,** Garlock E, Allison MJ, Michaud R, Lo R, Round JM, Helbing CC, Verreault J, Brinkman FSL (2022). Investigating the relationship between the skin microbiome and flame retardant exposure of the endangered St. Lawrence Estuary beluga. Frontiers in Environmental Science. 10: 954060. DOI:10.3389/fenvs.2022.954060
4. Bedi S*, Richardson TM*, **Jia B*,** Saab H, Brinkman FSL, Westley M (2022). Similarities between bacterial GAD and human GAD65: Implications in gut mediated autoimmune type 1 diabetes. PloS one. 17(2): e0261103. DOI:10.1371/journal.pone.0261103
5. Maguire F*, **Jia B***, Gray KL, Lau WYV, Beiko RG, Brinkman FSL.(2020). Metagenome-assembled genome binning methods with short reads disproportionately fail for plasmids and genomic Islands. Microbial Genomics. 6(10) DOI:10.1099/mgen.0.000436
6. **Jia B**, Raphenya AR, Alcock B, Waglechner N, Guo P, Tsang KK, Lago BA, Dave BM, Pereira S, Sharma AN, Doshi S, Courtot M, Lo R, Williams LE, Frye JG, Elsayegh T, Sardar D, Westman EL, Pawlowski AC, Johnson TA, Brinkman FS, Wright GD, McArthur AG. (2017). CARD 2017: expansion and model-centric curation of the comprehensive antibiotic resistance database (>190 citations). Nucleic acids research. 45(D1): D566-D573. DOI:10.1093/nar/gkw1004